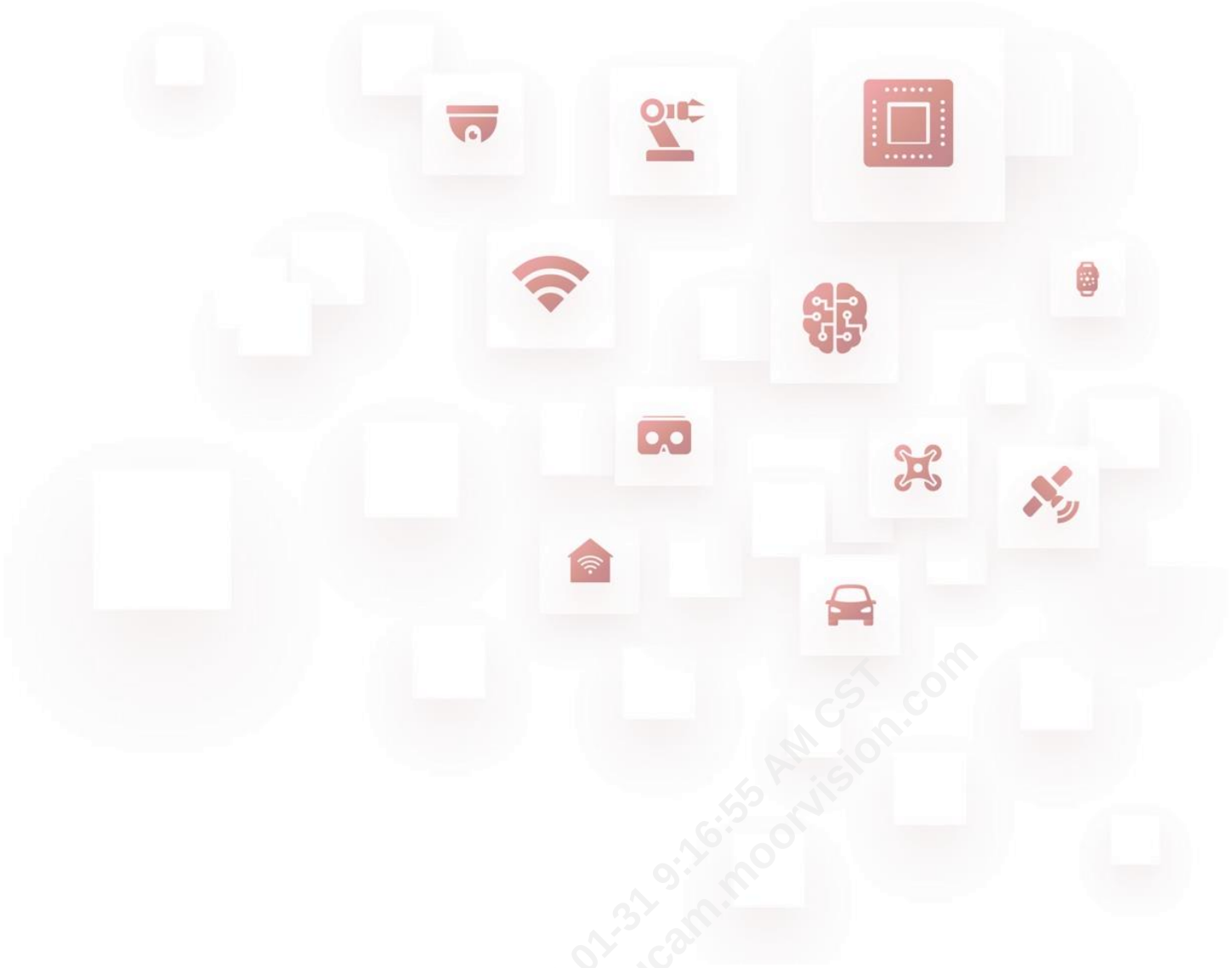
Thundercomm
AIOT ENABLER



2022-01-31 9:16:55 AM CST
Stefan@ucam.moorvision.com

Thundercomm TurboX C410/C610 Open Kit Linux Embedded

# Software Release Notes

Rev V1.0
Oct 31, 2022

DN: tc-gle-2410
SDK Version: LE2.0.CS.r002001

# Revision History

| Revision | Date | Description |
|---|---|---|
| 1.0 | Oct 31, 2022 | Initial release. |

Thundercomm Confidential
2025-01-31 9:16:55 AM CST
Stefan_ucam.moorvision.com

Copyright© 2022 Thundercomm Technology Co., Ltd. All rights reserved.

# Table List

*Table 3-1. LE SDK information*

*Table 5-1. Test report*

*Table 6-1. List of known issues*

Thundercomm Confidential
2025-01-31 9:16:55 AM CST
Stefan_ucam.moorvision.com

Copyright© 2022 Thundercomm Technology Co., Ltd. All rights reserved.

# About This Document

- Illustrations in this documentation might look different from your product.
- Depending on the model, some optional accessories, features, and software programs might not be available on your device.
- Depending on the version of operating systems and programs, some user interface instructions might not be applicable to your device.
- Documentation content is subject to change without notice. Thundercomm makes constant improvements on the documentation of the products, including this guidebook.
- Function declarations, function names, type declarations, attributes, and code samples appear in a different format, for example, `cp armcc armcpp`.
- Code variables appear in angle brackets, for example, `<number>`.
- Button, tool, and key names appear in bold font, for example, click **Save** or press **Enter**.
- Commands to be entered appear in a different font; on the host computer use $ as shell prompt, while on the target device use # as shell prompt, for example,

```
$ adb devices
# logcat
```

- Part of the code that does not contain instructions appear in a different format, for example,

```
SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0777", GROUP="adm"
```

- Folders and files are formatted in italic, for example, *turbox_flash_flat.sh*.

Thundercomm Confidential
2025-01-31 9:16:55 AM CST
Stefan_ucam.moorvision.com

Copyright© 2022 Thundercomm Technology Co., Ltd. All rights reserved.

# Table of Contents

**Chapter 1. Introduction ..... - 2 -**

**Chapter 2. Development Environment Setup ..... - 3 -**

**Chapter 3. Get Started with LE SDK ..... - 4 -**

3.1. LE SDK information ..... - 4 -

3.2. Block diagram ..... - 4 -

3.3. SDK source tree structure ..... - 5 -

3.4. Download LE SDK ..... - 6 -

3.5. SDK version update ..... - 7 -

3.6. Build image ..... - 7 -

3.7. Pack ROM package ..... - 8 -

3.8. Flash images to board ..... - 9 -

**Chapter 4. Features ..... - 13 -**

4.1. Presented features ..... - 13 -

4.2. New features ..... - 15 -

4.3. On-going features ..... - 15 -

**Chapter 5. Test Report ..... - 16 -**

**Chapter 6. Known Issues ..... - 18 -**

**Appendix 1. Notices ..... - 19 -**

**Appendix 2. Trademarks ..... - 21 -**

Thundercomm Confidential
2025-01-31 9:16:55 AM CST
Stefan_ucam.moorvision.com

# Chapter 1. Introduction

This document provides information on detailed steps to obtain, build, and program QCS610 LE (Linux Embedded) SDK (Software Development Kit) into the TurboX C410/C610 open kit.

➲ **NOTE:** "TurboX" referred to herein is the English text of our registered trademark TURBOX.

The LE SDK supports the Linux operating system on the Thundercomm TurboX SOM. The LE SDK aims to provide a simple and flexible development platform to rapidly build IoT (Internet of Things) products with the QTI chipsets. The LE SDK provides the build framework, software components, tools, and sample applications to evaluate the platform and build software images for the target product.

The LE SDK is based on the Yocto Project 2.6 release and follows several of its development principles. The Yocto Project is used by the LE SDK to build framework.

Thundercomm Confidential
2025-01-31 9:16:55 AM CST
Stefan_ucam.moorvision.com

# Chapter 2. Development Environment Setup

Refer to *TurboX SOM SDK Manager User Guide* to set up the development environment.

➲ **NOTE:** Make sure that you use the sdkmanager built based on Ubuntu 18.04. When you start the sdkmanager, the following window will appear.

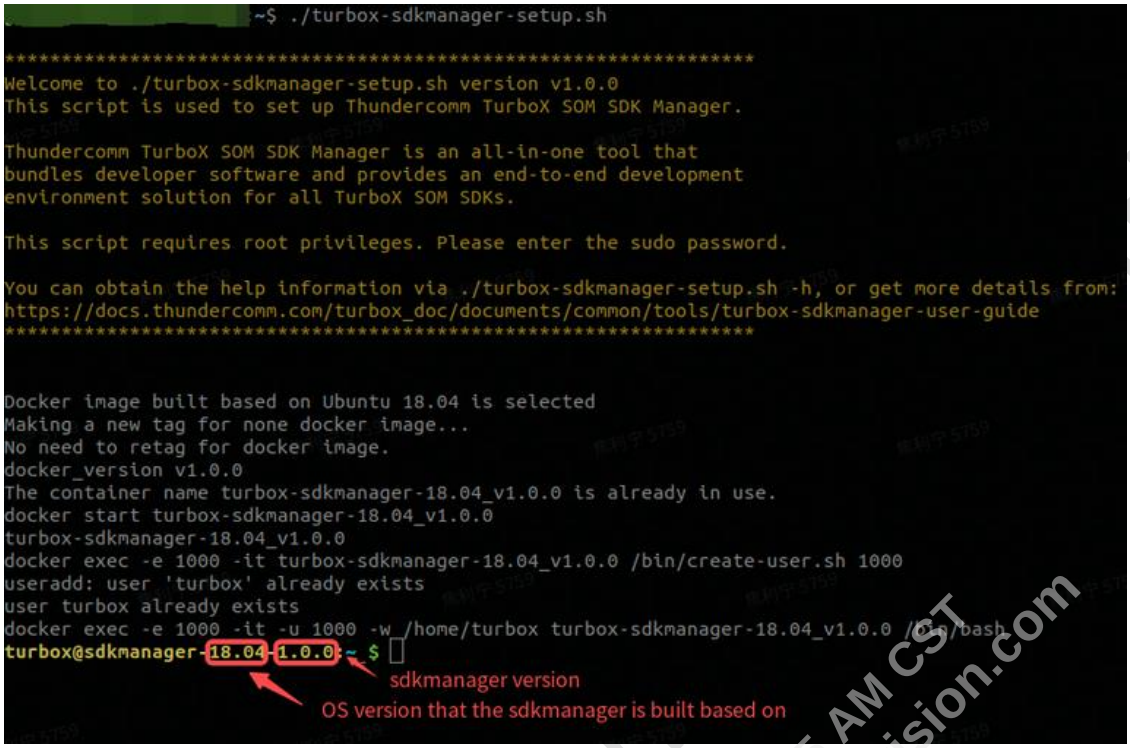



*Figure 2-1.sdkmanager Window*

Thundercomm Confidential
2025-01-31 9:16:55 AM CST
Stefan_ucam.moorvision.com

# Chapter 3. Get Started with LE SDK

## 3.1. LE SDK information

***Table 3-1. LE SDK information***

| SDK Version | Baseline | Baseline Distribution | OS Version |
|---|---|---|---|
| LE2.0.CS.r002001 | QCS610.LE.2.0 | Post-CS[1]0.0.007.2 | Linux Embedded 2.0 |

## 3.2. Block diagram

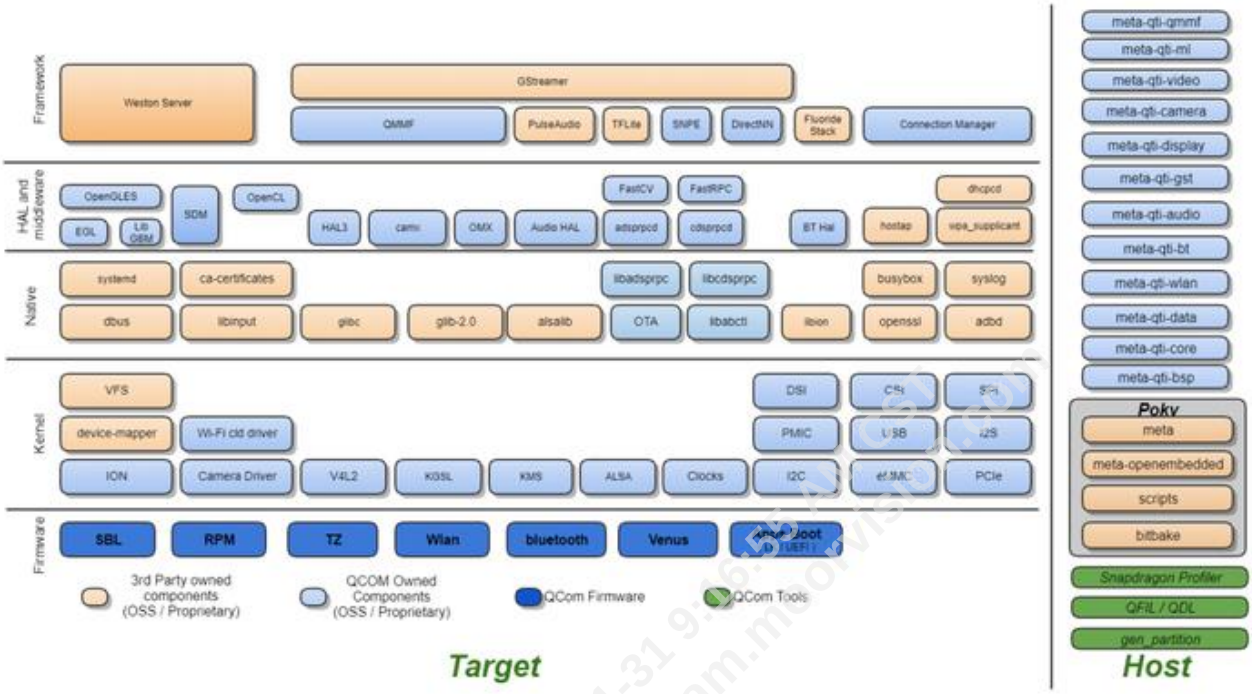



*Figure 3-1.Block Diagram*

Thundercomm Confidential 2025-01-31 9:16:53 Stefan_ucam.mcorvisionai

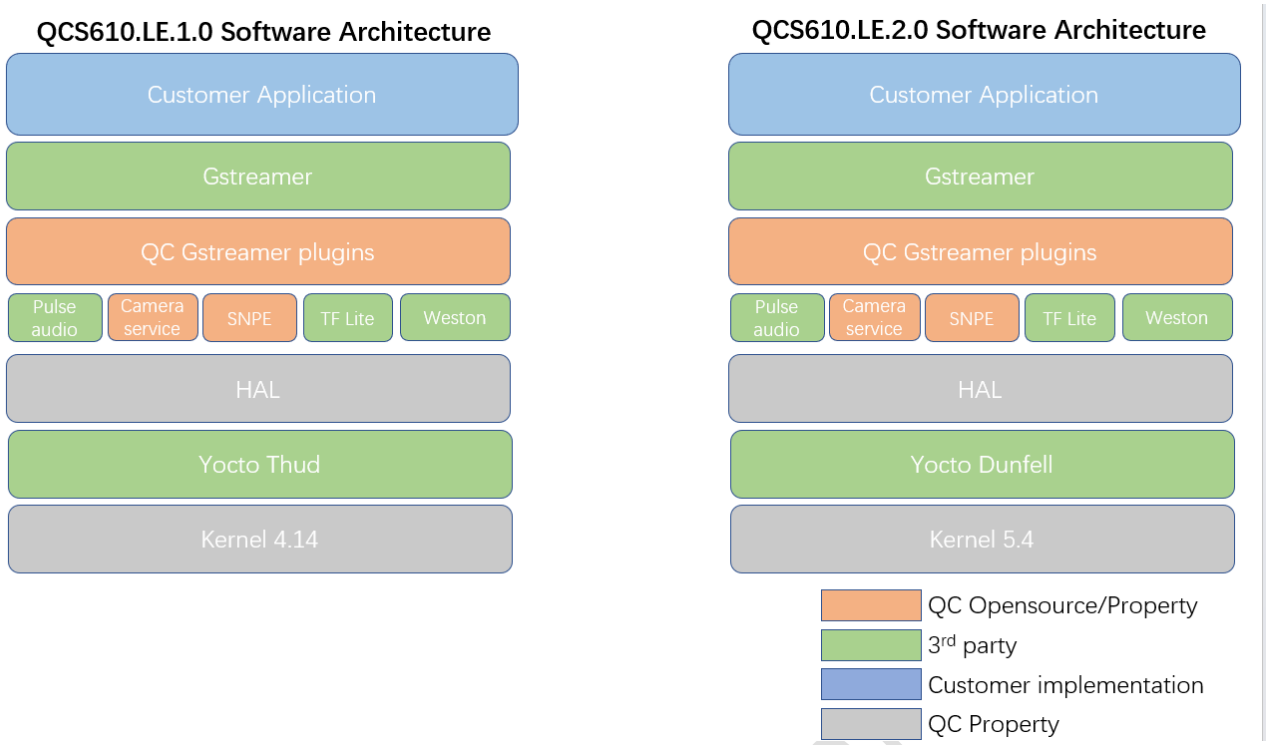



*Figure 3-2.Difference Between QCS610 LE1.0 & QCS610 LE2.0*

## 3.3. SDK source tree structure

```
SDK.Turbox_C610.LE2.0_CS/
├── about.html -> common/about.html      //SDK version info(Product,Distribution, …)
├── adsp_proc      //aDSP image is released as a binary and no build compilation is needed
├── aop_proc       //AOP image is released as a binary and no build compilation is needed
├── apps_proc
   ├── build-qti-distro-fullstack-virtualization-debug   /*Directory of temporary file and
image files generated by compilation */
   ├── disregard
   ├── downloads                    //Software packages required for compilation
   ├── poky                         //Yocto's recipe directory
   ├── prebuilt_HY11                //Qcom privatized software package
   ├── setup-environment -> poky/qti-conf/set_bb_env.sh     //Script for set the build
environement
   ├── src         //HLOS source code, such as linux kernel, filesystem, vendor, OTA, etc.
   ├── sstate-cache
   └── syncbuild.sh
├── boot_images  //BOOT image is released as a binary and no build compilation is needed
├── BP-CODE        //non-HLOS binary files
├── btfm_proc      //BTFM image is released as a binary and no build compilation is needed
├── cdsp_proc      //cDSP image is released as a binary and no build compilation is needed
├── common/config/emmc/partition.xml          //eMMC partition file
├── contents.xml -> common/contents.xml       //image contents file
├── modem_proc    //MODEM image is released as a binary and no build compilation is needed
├── trustzone_images //TZ image is released as a binary and no build compilation is needed
├── turbox
│   └── tools
│       ├── build_script                      //SDK build scripts
│       └── flash_tools                       //Flash tools files(qdl, …)
├── turbox_build.sh -> turbox/tools/build_script/common/turbox_build.sh //SDK build script
├── turbox_flash.sh -> turbox/tools/build_script/device/turbox_flash.sh //Fastboot flash
script
├── venus_proc   //VIDEO image is released as a binary and no build compilation is needed
├── wdsp_proc    //wDSP image is released as a binary and no build compilation is needed
└── wlan_proc    //WLAN image is released as a binary and no build compilation is needed
```

## 3.4. Download LE SDK

Follow the steps below to activate an account and download the SDK code.

### 3.4.1. Configure git global variable

Configure user name and email.

```
turbox@sdkmanager-18.04-1.2.0:~ $ git config --global user.email "<User Email>"
turbox@sdkmanager-18.04-1.2.0:~ $ git config --global user.name "<User Name>"
```

➲ **NOTE:**

- Please use the email and user name you have registered at *Thundercomm website* to replace `<User Email>` and `<User Name>` in the above commands.
- Do not enter the angle brackets "`<>`" which are used for indicating code variables.

### 3.4.2. Account activation

Add the account/password information to *$HOME/.netrc* by running the following command.

```
turbox@sdkmanager-18.04-1.2.0:~ $ echo "machine partner.thundercomm.com login <User Email>
password <password>" > ~/.netrc
```

➲ **NOTE:** Please use the email/password you have registered at *Thundercomm website* to replace `<User Email>` and `<password>` in the above command.

### 3.4.3. Download the SDK code

➲ **NOTE:** Make sure that you download the LE SDK code under *~/workspace* or its subdirectory.

Step 1.Initialize the repository.

```
turbox@sdkmanager-18.04-1.2.0:~ $ repo init -u
https://partner.thundercomm.com/sdkgit/a/manifest -b SDK.Turbox-QCS610.LE.2.0 -m
turbox-c610-le2.0-dev.release.CS.r002001.xml --repo-
url=https://partner.thundercomm.com/sdkgit/a/tools/repo --repo-branch=stable --no-
repo-verify
```

Step 2.Synchronize the source code.

```
turbox@sdkmanager-18.04-1.2.0:~ $ repo sync -c -j4 --no-tags
```

Thundercomm Confidential

## 3.5. SDK version update

If you have downloaded the old-version SDK, there are two ways to download the new SDK fast.

**Option 1. Upgrade the local old folder to a new release**

Navigate to the old local code folder. You can run the repo init command of new release as specified in *3.4.3. Download the SDK code*, then run `repo sync -j4 -c --no-tags` to receive the new code.

**Option 2. Download the new SDK source code to a new folder (recommended)**

Create a new folder, and run `repo init` with `-reference` value.

For example:

```
turbox@sdkmanager-18.04-1.2.0:~ $ repo init -u
https://partner.thundercomm.com/sdkgit/a/manifest -b SDK.Turbox_C610.LE2.0 -m
turbox-c610-le2.0-dev.release.CS.r002001.xml --repo-
url=https://partner.thundercomm.com/sdkgit/a/tools/repo --repo-branch=stable --
no-repo-verify --reference=<old_sdk_code_path>
$ repo sync -j4 -c --no-tags
```

➲ **NOTE:** In `--reference=<old_sdk_code_path>`, the `old_sdk_code_path` is the absolute path of your old SDK folder.

## 3.6. Build image

To compile the HLOS, run the following script file according to your image version in the root directory.

➲ **NOTE:** Please install the necessary compilation tools:

```
turbox@sdkmanager-18.04-1.2.0:~ $sudo apt-get install uuid-dev
```

- Build debug version:

```
turbox@sdkmanager-18.04-1.2.0:~ $ ./turbox_build.sh -a -l -v debug
```

- Build perf version:

```
turbox@sdkmanager-18.04-1.2.0:~ $ ./turbox_build.sh -a -l -v perf
```

➲ **NOTE**: UART debug cannot be used in perf version.

- Run the following commands to rebuild *kernel* additionally (if necessary).

```
$ cd apps_proc
$ MACHINE=qcs610-odk-64 DISTRO=qti-distro-fullstack-debug
$ source poky/qti-conf/set_bb_env.sh
$ bitbake -fc do_makeboot qti-multimedia-image
```

➲ **NOTE**: The default *boot.img* has verify enabled and cannot be flashed unless the kernel is rebuilt. Hence, when rebuilding only the kernel, you need to flash the *boot.img.noverity* which has verify disabled, refer to *turbox_flash.sh* to flash the *boot.img.noverity*.

- Run the following commands to rebuild *dtbo* additionally (if necessary).

```
$ cd apps_proc
$ MACHINE=qcs610-odk-64 DISTRO=qti-distro-fullstack-debug
$ source poky/qti-conf/set_bb_env.sh
$ bitbake -c cleanall linux-msm && bitbake linux-msm
```

Thundercomm Confidential

## 3.7. Pack ROM package

Step 1. Run the following command to update common information according to your image version.

- **Debug version:**

```
turbox@sdkmanager-18.04-1.2.0:~ $  ./turbox_build.sh -u --uci-imf -v debug
```

- **Perf version:**

```
turbox@sdkmanager-18.04-1.2.0:~ $ ./turbox_build.sh -u --uci-imf -v perf
```

➲ **NOTE:** The command is successfully executed with the following log displayed.

```
- update_common_info.py:======  UPDATE COMMON INFO COMPLETE=======
```

Step 2. Run the following command to pack the ROM package. You can find the packed ROM package at *turbox/output*. Replace `<debug/perf>` with debug or perf.

- Pack Fastboot image

```
turbox@sdkmanager-18.04-1.2.0:~ $ ./turbox_build.sh --zip_fastboot_build -l -v
<debug/perf>
```

- Pack FlatBuild image

```
turbox@sdkmanager-18.04-1.2.0:~ $ ./turbox_build.sh --zip_flat_build -l -v
<debug/perf>
```

➲ **NOTE:** Customers can also get the ROM package provided by Thundercomm from:

- **For debug version:**
  - *FlatBuild_TurboX_C610_xx.xx_LE2.0.R.debug.CS.r002001.zip*

    md5: 1555d457e2a526cefb56f0b0b2b1ea5a
  - *FastbootBuild_TurboX_C610_xx.xx_LE2.0.R.debug.CS.r002001.zip*

    md5: 10d066cf8375d5526c68cbcdc7fe5eb7
- **For perf version:**
  - *FlatBuild_TurboX_C610_xx.xx_LE2.0.R.perf.CS.r002001.zip*

    md5: dcf0a104d09dd5199e93e8b4601145d4
  - *FastbootBuild_TurboX_C610_xx.xx_LE2.0.R.perf.CS.r002001.zip*

    md5: 0dd47f44d63c30eaf8e0197401d32106

Thundercomm Confidential

## 3.8. Flash images to board

➲ **NOTE**: If the device fails to boot up after image flash with **Fastboot**, try to flash the **FlatBuild** image again with *turbox_flash_flat.sh or TurboX Easy Flash*.

### 3.8.1. Flash images on Windows

You can flash images on Windows by using TurboX Easy Flash, which is a tool created by Thundercomm to help SOM developers easily flash device image, backup and restore data, etc.

For tool acquisition and operation procedures, refer to *TurboX Easy Flash User Guide*.

### 3.8.2. Flash images on Linux

- **Flash images on Linux with Fastboot**

  Step 1. Enter the **Fastboot** mode.

  Option 1. Enter the following command when the device in **ADB** mode.

```
$ adb root
$ adb reboot bootloader
```

  Option 2. Power off the device, hold the **Volume down** button and click the **Power on** button, then release the **Volume down** button about 5 seconds later.

  ➲ **NOTE**: To check if the device has entered the **Fastboot** mode, run the following command:

```
$ sudo fastboot devices
```

  Successful with the following output appearing on the shell.

```
21f10525   fastboot
```

  Step 2. Flash images.

  Execute the commands below to flash images to the board. Replace `<folder name of image package>` with the actual image package name of your product.

```
$ cd <folder name of image package>
$ sudo ./turbox_flash.sh --all -r -f -v <ROM type>
```

Thundercomm Confidential

➲ **NOTE:**

- Please replace **<ROM type>** with **<debug>** (for debug version) or **<perf>** (for perf version) as required. Do not enter the angle brackets "**<>**" which are used for indicating code variables.
- Enter **./turbox_flash.sh -h** to acquire detailed usage.

```
$ ./turbox_flash.sh -h
Usage:
  bash ./turbox_flash.sh [options] [options]
  Images except gpt,userdata,persist will be flashed if no options input

Description:
  flash images with fasboot

Options:
  -h, --help   display this help message
  -v, --variant <debug/perf/user>    set the build variant.(default: debug)
  -f, --force  forcibly flash gpt,persist and userdata without confirmation.
             User data and calibration data(BT/WIFI MAC address, etc..) will
be ereased.
  -r, --reboot     reboot the device automatically after flashing completes.
  -b, --nonhlos     flash all bp images(xbl,aop,ftfm,tz,dsp,modem,pmic).
  -a, --hlos       flash all ap
images(aboot,bootimage,recovery,system,vendor,cache,mdtp,userdata,persist).
  --all       flash all images

Example:
  ./turbox_flash.sh
     Flash the images except gpt,userdata,persist.

  ./turbox_flash.sh --force
     Flash all images.

  ./turbox_flash.sh -hlos
Flash all ap images.
```

Thundercomm 2025-01-14 9:16:55 Alex.csf stefan_ucan@moorvision.com

- **Flash images on sdkmanager with *turbox_flash_flat.sh* tool**

  ➲ **NOTE:** Please update the sdkmanager version to v1.2.0 which has built-in *turbox_flash_flat.sh*.

  Step 1.Stop the ModemManager on your host PC and close the application that occupies the ttyUSB0 port, such as **Minicom**.

  ```
  $ sudo systemctl stop ModemManager
  ```

  Step 2.Prepare the FlatBuild images.

  Check if the images are ready and exists in sdkmanager, then enter the image folder. Replace with the actual image package name of your product.

  ```
  turbox@sdkmanager-18.04-1.2.0:~/workspace $ cd <folder name of image package>
  ```

  Step 3.Enter 9008 Mode.

  1) Power off the device by disconnecting the power cable and USB cable.
  2) Connect the power cable.
  3) Hold the **Force USB Boot** button.
  4) Press the **Power on** button and wait for about 3 seconds.
  5) Release the **Force USB Boot** button and **Power on** button.
  6) Connect the Type-C USB cable to an Ubuntu computer.

  To check if your device has entered the **9008 Mode**, run the following command:

  ```
  turbox@sdkmanager-18.04-1.2.0:~ $ lsusb
  ……
  Bus 001 Device 042: ID 05c6:9008 Qualcomm, Inc. Gobi Wireless Modem (QDL mode)
  …… .
  ```

Thundercomm Confidential
2025-01-31 9:16:55 AM CST
Stefan_ucam.mo@vision.com

Step 4.Start to flash the FlatBuild images.

Run the following command under the image folder.

```
turbox@sdkmanager-18.04-1.2.0:~/workspace/<folder name of image package> $ turbox-flash-flat.sh
```

If the flashing process is successful, the following log will be printed.

```
11:20:09: INFO:      _            (done)
11:20:09: INFO:     | |
11:20:09: INFO:   __| | ___  _ __   ___
11:20:09: INFO:  / _` |/ _ \| '_ \ / _ \
11:20:09: INFO: | (_| | (_) | | | |  __/
11:20:09: INFO:  \__,_|\___/|_| |_|\___|
11:20:09: INFO: {All Finished Successfully}

11:20:09: INFO: Overall to target 30.119 seconds (85.78 MBps)
11:20:09: INFO: {percent files transferred 100.00%}
Writing log to '/home/turbox/workspace/FlatBuild_Turbox-C610_xx.xx_le2.0.D.debug.20221027.2351/port_trace.txt', might take a minute
```

The device board will automatically reboot.

Connect the USB 3.1 Gen 1 Type-C cable and check the adb device with the following commands.

```
$ adb devices
List of devices attached
ac55dc38        device
```

➲ **NOTE:** For more information, please go to: *Turbox_flash_flat-user-guide*.

# Chapter 4. Features

This chapter introduces the software features supported by the DK.

## 4.1. Presented features

This section describes the key software features that have been included in earlier versions.

- **SDK**
  - OpenEmbedded Linux 2.0
  - Yocto Dunfell(v3.1)
  - GCC toolchain (v8.2.0)
  - glibc (v2.28)
  - systemd (v1.239)
- **Platform**
  - Linux Kernel 5.4
  - 64-bit only
  - UBWC
  - OTA
  - HY22
  - DTB Overlay
- **CV**
  - FastCV
- **Security**
  - DM-Verity
  - SELinux
  - KASan
  - FDE
  - VBLE
  - Non-Root Processes
- **Connectivity**
  - WLAN 802.11ac
  - USB 3.1 Gen 1
  - Bluetooth 5.0
  - WPA-R3
- **Storage**
  - eMMC
  - UFS 2.0

Thundercomm Confidential
2025-01-31 9:16:55 AM CST
Stefan_ucam.moorvision.com

- **Multimedia**
  - Weston 8.0
  - GStreamer (v1.16) Framework
  - Hexagon-SDK
- **Camera**
  - Staggered HDR (v2&v3.8)
  - EIS (v2.0)
  - IRLED
  - LDC and EIS
  - Dual Independent Cameras
  - Force sensor mode to bypass default logic
  - Manual disablement of IR cut filter with IR LED OFF
  - Multiple black level
- **Video**
  - JPEG DMA Engine
  - H264 & HEVC encode and decode
- **Audio**
  - AAC, PCM and MP3 (only playback) Encode and Playback
  - Audio + Hands Free Profile concurrency
  - Voice UI 3.0
- **Display**
  - DP
  - LCD
- **Graphics**
  - OpenGL ES 3.2
  - OpenCL 2.1
- **AI**
  - BENCHMARK APP
  - AI Inferencing via SNPE v1.36
  - AI Inferencing via TFLite 2.0
  - TFLite supports NNAPI delegate offloading graph processing to DSP
  - SNPE run on GPU CPU and DSP
  - TF Lite DSP run-time support through HexNN
- **Battery**
  - Software-based fuel gauge solution for battery powered devices

Thundercomm Confidential
2025-01-31 9:16:55 AM CST
Stefan_ucam.moorvision.com

## 4.2. New features

- Make bootup changes to improve cold boot latency for IoT
- Support dual cameras via VC over single CSI and up to 4 cameras over dual CSIs
- Support dm-verity with SquashFS
- Support OTA for SquashFS system image
- Add UFS 3.1 driver support to QCS610 with data rate matching UFS 2.0
- Add GST plugin to support generic tensor in and out
- Add GST plugin to perform multiple cropping and resizing per frame
- Support new screen (VIEWe UE055HD-AP40-A047D)
- Support IMX334 camera module

## 4.3. On-going features

- HDMI-IN LT6911UXC bring-up
- BT/Wi-Fi/59/Red/Green/Blue LED bring-up
- LSM6DSOTR sensor bring-up

Thundercomm Confidential
2025-01-31 9:16:55 AM CST
Stefan_ucam.moorvision.com

# Chapter 5. Test Report

***Table 5-1. Test report***

| Test Case | Result |
|---|---|
| FlatBuild version flash | Pass |
| Fastboot version flash | Pass |
| Metabuild version flash | Pass |
| Boot_adb reboot | Pass |
| Boot_adb reboot bootloader | Pass |
| Boot_fastboot reboot | Pass |
| Fastboot (adb & volume down) | Pass |
| Power on button | Pass |
| Volume up/down button | Pass |
| Force USB Boot button | Pass |
| Memory Information | Pass |
| Power suspend | Fail |
| 64-bit user space | Pass |
| Disable verity | Pass |
| SD Card auto mount/unmount/multiple plug & unplug | Pass |
| SD card file push/pull | Pass |
| Multimedia_GST_recorder_4k_30_AVC_MP4_mux | Pass |
| Multimedia_GST_recorder_4k_30_HEVC_MP4_mux | Pass |
| Multimedia_GST_recorder_1080p_30_AVC_filesave | Pass |
| Multimedia_GST_Snapshot_One 1080p AVC mp4 muxed video with JPEG snapshots | Pass |
| Multimedia _GST_Overlay_1080p_30 YUV with UserText overlay | Pass |
| Multimedia _GST_Display_1080p_30 Live preview on a local display | Pass |
| Multimedia _GST_Display_1080p_30 Live preview over TCP/RTSP | Pass |
| Camera_LDC/ZOOM | Pass |
| Camera_Dual Independent Cameras (IMX577) | Pass |
| Audio Line IN | Pass |
| Audio_1-6 D-MIC recorder individually | Pass |
| Audio_1-6 channels recorder concurrency with GST | Pass |
| Audio_1-6 channels recorder concurrency with parecord | Pass |
| Audio headphone | Pass |
| Audio speaker | Pass |
| Display_DP | Pass |
| Display_LCD | Pass |

Thundercomm Confidential

| Test Case | Result |
|---|---|
| Display_Weston color Bar | Pass |
| USB_Type-C functionality | Pass |
| USB_Type-A functionality | Pass |
| USB_Micro for UART | Pass |
| Ethernet_RJ45 connect | Pass |
| Ethernet_ Ping test | Pass |
| Connectivity_Bluetooth | Pass |
| Connectivity_ Wi-Fi_connect to AP | Pass |
| Battery | N/A |
| Charge the battery via Type-c | N/A |
| 12V DC power supply | Pass |
| AI inferencing via TFLite 2.0 | Pass |

Thundercomm Confidential
2025-01-31 9:16:55 AM CST
Stefan_ucam.moorvision.com

# Chapter 6. Known Issues

***Table 6-1. List of known issues***

| Issue No. | Description | Status |
|---|---|---|
| Limitations-002 | RJ45 indicator is abnormal. | Open |
| BUG-13155 | System cannot enter Sleep Mode. | Ongoing |
| BUG-13253 | Ethernet RGMII cannot work in 10M/s and 100M/s modes. | Ongoing |
| BUG-13254 | The new display touch panel is unstable. | Ongoing |
| BUG-Q-3147869 | [QCS610.LE.2.0][UEFI]: Charger test fails with ERR 3. | Ongoing |
| BUG-Q-3250714 | [QCS610.LE.2.0][C610+IMX334]: “Segmentation fault” appears during the switchover between ready state and playing state in storage use case. | Ongoing |

Thundercomm Confi
2025-01-31 9:16:55 AM CST
Stefan_ucam.moorvision.com

# Appendix 1. Notices

Thundercomm may have patents or pending patent programs covering subject matter described in this document. The furnishing of this document does not give you any license to these patents. You can send license inquiries to service@thundercomm.com.

THUNDERCOMM PROVIDES THIS PUBLICATION "AS IS" WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESS OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF NON-INFRINGEMENT, MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE. Some jurisdictions do not allow disclaimer of express or implied warranties in certain transactions; therefore, this statement may not apply to you.

Changes are made periodically to the information herein; these changes will be incorporated in new editions of the publication. To provide better service, Thundercomm reserves the right to improve and/or modify the products and software programs described in the manuals, and the content of the manual, at any time without additional notice.

The software interface and function and hardware configuration described in the manuals included with your development board or system on module might not match exactly the actual configuration of that you have purchased. For the configuration of the product, refer to the related contract (if any) or product packing list, or consult the distributor for the product sales. Thundercomm may use or distribute any of the information you supply in any way it believes appropriate without incurring any obligation to you.

The products described in this document are not intended for use in implantation or other life support applications where malfunction may result in injury or death to persons. The information contained in this document does not affect or change Thundercomm product specifications or warranties. Nothing in this document shall operate as an express or implied license or indemnity under the intellectual property rights of Thundercomm or third parties. All information contained in this document was obtained in specific environments and is presented as an illustration. The result obtained in other operating environments may vary.

The information of this document should not be as any invitation for offer or any advice to the visitors. Please consult the professional comments from the sales consultant prior to do any actions of investment or purchase.

Thundercomm may use or distribute any of the information you supply in any way it believes appropriate without incurring any obligation to you.

Any references in this publication to non-Thundercomm Web sites are provided for convenience only and do not in any manner serve as an endorsement of those Web sites. The materials at those Web sites are not part of the materials for this Thundercomm product, and use of those Web sites is at your own risk. Thundercomm shall not be responsible for the content of the third party.

Any performance data contained herein was determined in a controlled environment. Therefore, the result obtained in other operating environments may vary significantly. Some measurements may have been made on development-level systems and there is no guarantee that these measurements will be the same on generally available systems. Furthermore, some measurements may have been estimated through extrapolation. Actual results may vary. Users of this document should verify the applicable data for their specific environment.

This document is copyrighted by Thundercomm and the property right of the date mentioned in this document, including but not limited trademarks, patents, copyrights, trade name etc. are not covered by any open-source license. Thundercomm may update this document at any time without notice.

Anyone doesn’t have the right to amend, reprint, republication, reproduce, transmit, distribute or any other way to use this document in business or public purpose without the prior written consent by Thundercomm.

E-mail messages sent to Thundercomm via the Internet are not guaranteed to be completely secure.Thundercomm shall not be liable for any loss incurred by the surfer when transmitting any information

Thundercomm Confidential

over the Internet or for any loss incurred by Thundercomm when sending any information over the Internet at your request.

Thundercomm has all rights under other relevant exemptions provided by laws and regulations, and Thundercomm's failure to claim or delay in claiming such rights shall not be deemed to be a waiver of such rights by Thundercomm.

Thundercomm reserves the right of final interpretation of this document.

Thundercomm Confidential
2025-01-31 9:16:55 AM CST
Stefan_ucam.moorvision.com

# Appendix 2. Trademarks

Thundercomm, Thundercomm TurboX, TURBOX, Thundersoft turbox are trademarks of Thundercomm Corporation or its associate companies in China and/or other countries. Intel, Intel SpeedStep, Optane, and Thunderbolt are trademarks of Intel Corporation or its subsidiaries in the U.S. and/or other countries. Microsoft, Windows, Direct3D, BitLocker, and Cortana are trademarks of the Microsoft group of companies. Mini DisplayPort (mDP), DisplayPort, and VESA are trademarks of the Video Electronics Standards Association. The terms HDMI and HDMI High-Definition Multimedia Interface are trademarks or registered trademarks of HDMI Licensing LLC in the United States and other countries. Wi-Fi, Wi-Fi Alliance, WiGig, and Miracast are registered trademarks of Wi-Fi Alliance. USB-C is a registered trademark of USB Implementers Forum. All other trademarks are the property of their respective owners.

Thundercomm Confidential
2025-01-31 9:16:55 AM CST
Stefan_ucam.moorvision.com